Worksheets - 15%
Quizzes - 20%
Mid Sem - 30%
End Sem - 35%

Sample space: set of all possible outcomes of an experiment.

Event: subset of the sample space

Naive def^n of probability:

$$P(A) = \frac{\#\text{ favourable outcome}}{\#\text{ possible outcome}}$$

HH
HT
TH
TT

Assumptions: (i) all outcomes are equally likely

(ii) finite sample space

Basic principles of counting:

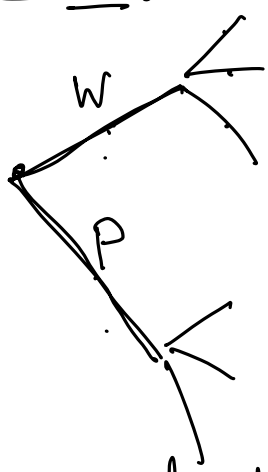
multiplication rule:

If we have an experiment with n_1 outcomes for each outcome in the 1st experiment, we have n_2 many outcome

..... n_r outcome for rth experiment.

Then $n_1 n_2 \ldots n_r$ possible outcome

Tree diagram

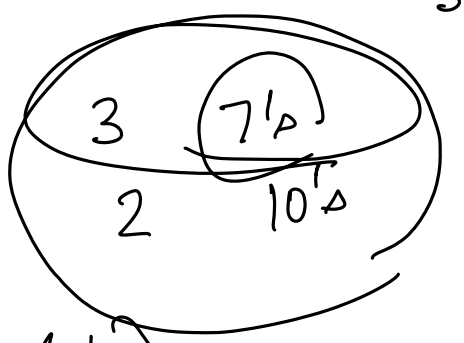


$6 = 2 \cdot 3$

$3 \cdot 2$

Ex. Find probability of full house in poker

5 cards



$$\frac{13 \cdot \binom{4}{3} \cdot 12 \cdot \binom{4}{2}}{\binom{52}{5}}$$

Sampling table :

Choose k objects out of n

	order matters	order does not matter
with replacement	n^k	*
without replacement	$n(n-1)(n-2)\cdots(n-k+1)$	^nC_k

Ex. Try to think about *.

$$\binom{n+k-1}{k}$$

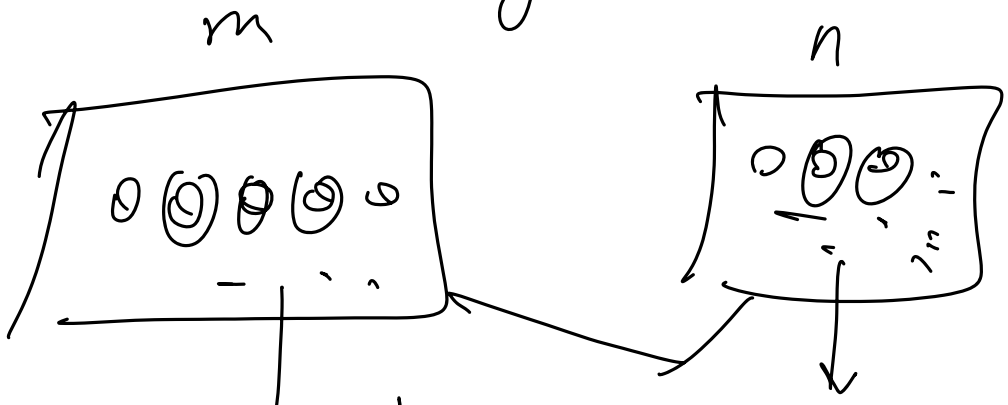
Proof by interpretation

$$n\binom{n-1}{k-1} = k\binom{n}{k}$$

Choose k from n with 1 person designated as President

$$① \quad \binom{m+n}{k} = \sum_{j=0}^{k} \binom{m}{j}\binom{n}{k-j}$$

Vandermonde's identity



$$\binom{m+n}{k} = \sum_{j=0}^{k} \binom{m}{j}\binom{n}{k-j}$$

Non naive def.^{n} of probability

Probability space consists of S & P

$S \to$ sample space

$P \to$ Probability func

$A \subseteq S$ as input

$P(A) \in [0,1]$

(i) $P(\phi) = 0$; $P(S) = 1$

(ii) $A_1, A_2, \ldots\ldots$ are disjoint

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Inclusion - Exclusion

$$P(A_1 \cup A_2 \cup A_3 \cup \ldots \cup A_n)$$

$$= \sum_{i=1}^{n} P(A_i) - \sum_{i<j} P(A_i \cap A_j)$$

$$+ \sum_{i<j<k} P(A_i \cap A_j \cap A_k)$$

$$\ldots\ldots + (-1)^{n+1} P(A_1 \cap A_2 \ldots \cap A_n)$$

Matching problem (1713)

What is the probability that there is atleast 1 card whose position on the deck is same as the number written on it.

(Deck of n cards $\rightarrow$ labelled it as $1, 2, \ldots n$)

A_j = event that the card jth matches with no. j

$$P(A_1 \cup A_2 \cup \ldots \cup A_n)$$

$$P(A_j) = \frac{1}{n} = \frac{(n-1)!}{n!}$$

$$P(A_1 \cap A_2) = \frac{(n-2)!}{n!}$$

$$P(A_1 \cap A_2 \ldots \cap A_k) = \frac{(n-k)!}{n!}$$

Apply inclusion - exclusion

$$P(A_1 \cup \cdots \cup A_n) = n \cdot \frac{1}{n} - {}^nC_2 \cdot \frac{(n-2)!}{n!}$$

$$+ {}^nC_3 \frac{(n-3)!}{n!}$$

$$\cdots\cdots (-1)^{n+1} \cdot \frac{1}{n!}$$

$$= 1 - \frac{1}{2!} + \frac{1}{3!} - \cdots (-1)^{n+1} \cdot \frac{1}{n!}$$

as $n \to \infty$

$$\approx 1 - \frac{1}{e}$$

$$P(\text{no match}) = 1/e$$

Frequentist interpretation of probability

In the frequentist interpretation

$P(E)$: Proportion of times, in repeated & independent trials of an experiment, in which E occurs

let n = # times the process is performed

m = # times the outcome E happens

out of the 1st n trials

$$P(E) = \lim_{n \to \infty} \frac{m}{n}$$

Remarks: ① Frequentist perspective generalizes classical "

② Just a "thought" experiment in some cases

Subjective perspective:

Axiomatic perspective:

Independent events: A, B are independent if $P(A \cap B) = P(A)P(B)$

Ex. Experiment: Drawing 1 card at random from a deck of 52 cards

A: Card is an ace
B: Card is a heart

$P(A) = 4/52 = 1/13$; $P(B) = 13/52 = 1/4$

$$P(A \cap B) = 1/52$$